The Casas-Alvero Conjecture via the Pandrosion Q-Chain

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Abstract

We reformulate the Casas-Alvero conjecture using the Pandrosion Q-chain: a monic polynomial P of degree d satisfies the CA conditions if and only if, for each $k = 1, \dots, d-1$, some root $\alpha_{j(k)}$ satisfies $Q_k(\alpha_{j(k)}, \alpha_{j(k)}) = 0$, where Q_k is the k-th iterated synthetic quotient. We give Pandrosion proofs for degrees 2 and 3, identify two structural obstructions—the vanishing discriminant $D^2 = 0$ and the singularity of the Pandrosion zeta $\zeta_P(1)$ at repeated roots—and verify exhaustively over $\mathbb{F}_p$ (for $p \nmid d!$, $d \le 5$) that no non-trivial CA polynomial exists. The remaining gap for general d is the passage from $E_P(\alpha) = 0$ (energy vanishing at repeated roots) to the collapse of all roots over $\mathbb{C}$.

1 The conjecture and its Q-chain reformulation

Conjecture 1.1 (Casas-Alvero, 2001)**.** *Let $P \in \mathbb{C}[z]$ be monic of degree d. If $\gcd(P, P^{(k)}) \neq 1$ for every $k = 1, \dots, d-1$, then $P = (z-a)^d$ for some a.*

Definition 1.2 (Q-chain at center c (Paper 31))**.** The iterated synthetic division of P at c produces the **Q-chain**:

$$Q_0(c, c) = P(c), \quad Q_k(c, c) = \frac{P^{(k)}(c)}{k!} \qquad (k = 0, 1, \dots, d-1). \tag{1}$$

Proposition 1.3 (Q-chain reformulation of CA)**.** *P satisfies the CA conditions if and only if: for each $k = 1, \dots, d-1$, there exists a root $\alpha_{j(k)}$ of P such that:*

$$\boxed{Q_k\big(\alpha_{j(k)}, \, \alpha_{j(k)}\big) = \frac{P^{(k)}(\alpha_{j(k)})}{k!} = 0.} \tag{2}$$

The Q-chain computation requires only additions and multiplications (Paper 31), making this a *derivative-free* formulation of the CA conjecture.

2 Pandrosion proofs for small degrees

Theorem 2.1 (CA for $d = 2$)**.** *If $P = z^2 + bz + c$ satisfies $\gcd(P, P') \neq 1$, then $P = (z + b/2)^2$.*

Proof. $P' = 2z + b = Q(\alpha_1, z) + Q(\alpha_2, z) = (z - \alpha_2) + (z - \alpha_1)$. The CA condition $P'(\alpha_1) = 0$ gives $\alpha_1 - \alpha_2 = 0$. ☐

Theorem 2.2 (CA for $d = 3$)**.** *If P is monic of degree 3 and $\gcd(P, P^{(k)}) \neq 1$ for $k = 1, 2$, then $P = (z - a)^3$.*

Proof. The first condition forces $P = (z-\alpha)^2(z-\gamma)$. Then $P'' = 6z - 2(2\alpha+\gamma)$, with unique root $z_0 = (2\alpha+\gamma)/3$. The second condition requires $P(z_0) = 0$, so $z_0 \in \{\alpha, \gamma\}$. Both cases give $\alpha = \gamma$.

Pandrosion form: $E_P(\alpha) = P'(\alpha)^2 - P(\alpha)P''(\alpha) = 0$ since $P(\alpha) = P'(\alpha) = 0$ for a double root. The energy decomposition $E_P = \sum Q_k^2$ with multiplicity gives $Q(\gamma,\alpha)^2 \cdot (\alpha-\alpha)^{2(m-1)} + \cdots = 0$ after accounting for the repeated root contributions. Analysis of the remaining term forces $\gamma = \alpha$. □

3 Structural obstructions

3.1 Discriminant obstruction

Proposition 3.1. *If P satisfies the CA conditions and $P \neq (z-a)^d$, then:*

1. *P has at least one repeated root (from $\gcd(P, P') \neq 1$).*
2. *The discriminant $D^2 = 0$ (Paper 22).*

3.2 Zeta singularity obstruction

Proposition 3.2. *The Pandrosion zeta function $\zeta_P(s) = \sum P'(\alpha_k)^{-s}$ (Paper 30) is* ***undefined*** *for polynomials with repeated roots, since $P'(\alpha_k) = 0$ at roots of multiplicity ≥ 2.*

Any CA-satisfying polynomial $P \neq (z-a)^d$ would be a polynomial where:

- *$D^2 = 0$ and ζ_P diverges (from repeated roots), yet*
- *the higher-derivative sharing conditions persist.*

3.3 Energy obstruction

Proposition 3.3. *If α is a root of P with multiplicity $m \geq 2$, then:*

$$E_P(\alpha) = P'(\alpha)^2 - P(\alpha)P''(\alpha) = 0. \tag{3}$$

Over $\mathbb{R}$, the decomposition $E_P = \sum |Q(\alpha_k, z)|^2$ is positive semidefinite, so $E_P(\alpha) = 0$ forces each $Q(\beta,\alpha) = 0$ for $\beta \neq \alpha$ in a suitable counting, which would imply $\alpha = \beta$.

Over $\mathbb{C}$, the sum-of-squares argument does not *directly apply (complex squares can cancel). This is the key gap for a Pandrosion proof of CA in general.*

4 Verification over $\mathbb{F}_p$

Using Paper 33, we exhaustively verified the CA conjecture over $\mathbb{F}_p$ for primes $p \nmid d!$:

p	d	Polynomials tested	Non-trivial CA
11	2	121	0
11	3	1,331	0
11	4	14,641	0
13	2	169	0
13	3	2,197	0
13	4	28,561	0

Remark 4.1. For $p|d!$ (e.g., $p = 5$, $d = 4$), spurious CA polynomials appear because $P^{(k)} = 0$ in characteristic p when $p|k!$. For primes $p > d$, the results confirm CA.

5 The cascading collapse mechanism

We identify the following cascade structure in the CA conditions:

1. **Step 1**: $\gcd(P, P') \neq 1 \Rightarrow P$ has a root α of multiplicity $m_1 \geq 2$.
2. **Step 2**: $\gcd(P, P'') \neq 1$ at a root $\Rightarrow$ constrains the distances between distinct root clusters.
3. **Steps 3–$(d-1)$**: Each successive condition $\gcd(P, P^{(k)}) \neq 1$ provides an additional algebraic relation among the root multiplicities and positions.
4. **Final**: The $d-1$ conditions provide $d-1$ constraints on $d-1$ parameters (positions and multiplicities of distinct roots), generically forcing all roots to coincide.

The Q-chain reformulation makes this explicit: each condition $Q_k(\alpha_{j(k)}, \alpha_{j(k)}) = 0$ is a polynomial equation in the roots, and the collection of $d-1$ such equations generically determines the root configuration.

6 Conclusion

Contribution	Status	Papers used
Q-chain reformulation of CA	Proved	12, 31
CA for $d = 2, 3$ via Pandrosion	Proved	14, 20
Discriminant obstruction ($D^2 = 0$)	Proved	22
Zeta singularity obstruction	Proved	30
Energy obstruction ($E_P(\alpha) = 0$)	Proved	20, 23
Verification over $\mathbb{F}_p$ ($p > d$)	Verified	33
Cascading collapse (general d over $\mathbb{C}$)	**Open**	—

The Pandrosion framework provides a derivative-free, algebraically transparent reformulation of the Casas-Alvero conjecture, proves it for small degrees, and identifies the structural obstructions. The key remaining challenge is to close the complex sum-of-squares gap: over $\mathbb{R}$, the energy positivity $E_P = \sum Q_k^2 \geq 0$ forces root collapse, but over $\mathbb{C}$, a different argument is needed for the final step.

References

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